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Windows PowerShell
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PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W6D1_LangChain_Fundamentals> ollama list
NAME      ID              SIZE    MODIFIED
qwen2.5:3b 357c53fb659c    1.9 GB  4 hours ago
llama3.2:3b a80c4f17acd5    2.0 GB  5 hours ago
PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W6D1_LangChain_Fundamentals> python w6d1_langchain.py
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W6D1 TASK 1: LANGCHAIN CHAIN
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Input 1: What is machine learning?
Output: Machine learning (ML) is a subset of artificial intelligence that enables computers to automatically learn and improve from data, without being explicitly programmed. It involves training algorithms on large datasets to enable them to make predictions or decisions based on patterns and relationships in the data.
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Input 2: What is LangChain?
Output: LangChain is an open-source Python library that enables interoperability between blockchain networks and external data storage systems, such as AWS S3 or Google Cloud Storage. It facilitates data transfer and management across these platforms.
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Input 3: What is a vector database?
Output: A vector database is a type of data storage system designed for efficiently managing and querying large amounts of high-dimensional vector data, such as text documents, images, or audio signals. It stores vectors that represent these data points in a way that enables fast similarity searches, nearest neighbor queries, and other operations typically required in applications like search, recommendation, or object detection.
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Input 4: What is Python used for?
Output: Python is a versatile programming language used for:

1. Web development (e.g., Django, Flask)
2. Data analysis and science (e.g., NumPy, pandas, scikit-learn)
3. Artificial intelligence and machine learning (e.g., TensorFlow, Keras)
4. Automation and scripting
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4. Automation and scripting
5. Game development
6. Education and research

Its simplicity, flexibility, and extensive libraries make it a popular choice for many applications.
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Input 5: What is an API?
Output: An Application Programming Interface (API) is a set of defined rules and protocols that enables different software systems to communicate with each other, allowing data and functionality to be exchanged between applications.
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W6D1 TASK 2: CONVERSATION BUFFER MEMORY
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C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W6D1_LangChain_Fundamentals\w6d1_langchain.py:63: LangChainDeprecationWarning: The class 'ConversationBufferMemory' was deprecated in LangChain 0.3.1 and will be removed in 2.0.0. Use 'langchain.agents.create_agent' instead. For agents that need to remember prior interactions, use 'create_agent' with checkpointing or the 'Store' API. See https://docs.langchain.com/oss/python/langchain/short-term-memory and https://docs.langchain.com/oss/python/langchain/long-term-memory
memory = ConversationBufferMemory()

Turn 1
Human: My name is Sanjana.
AI: Hello Sanjana! It's nice to meet you. I don't have any prior knowledge from our conversation as this is the start of our interaction. How can I assist you today?
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Turn 2
Human: What is my name?
AI: Hello Sanjana, we just met and I recall that your name is indeed "Sanjana". You might want to ask me about anything else you need help with!
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Turn 3
Human: I am learning LangChain.
AI: That's great to hear, Sanjana! LangChain is a powerful open-source library for building conversational AI models. It provides a f
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AI: That's great to hear, Sanjana! LangChain is a powerful open-source library for building conversational AI models. It provides a flexible framework for creating custom interfaces and workflows. What specific aspects of LangChain are you looking to learn more about? Are you trying to integrate it into your project or build a new model from scratch?

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Turn 4
Human: What am I learning?
AI: You're learning LangChain, Sanjana! LangChain is an open-source library for building conversational AI models. It's designed to help developers create custom interfaces and workflows for their applications. You seem to be just starting out with it, so feel free to ask me any questions or clarify any concepts that you're unsure about. What specific areas of LangChain would you like to explore further?

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Turn 5
Human: Can you summarize what you know about me from this conversation?
AI: From our conversation, I know that your name is Sanjana. That's the only information I have about you so far. You've just started exploring LangChain, and we're in the process of getting to know each other better. How can I assist you further?

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STORED CONVERSATION HISTORY
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Human: My name is Sanjana.
AI: Hello Sanjana! It's nice to meet you. I don't have any prior knowledge from our conversation as this is the start of our interaction. How can I assist you today?
Human: What is my name?
AI: Hello Sanjana, we just met and I recall that your name is indeed "Sanjana". You might want to ask me about anything else you need help with!
Human: I am learning LangChain.
AI: That's great to hear, Sanjana! LangChain is a powerful open-source library for building conversational AI models. It provides a flexible framework for creating custom interfaces and workflows. What specific aspects of LangChain are you looking to learn more about? Are you trying to integrate it into your project or build a new model from scratch?
Human: What am I learning?
AI: You're learning LangChain, Sanjana! LangChain is an open-source library for building conversational AI models. It's designed to help developers create custom interfaces and workflows for their applications. You seem to be just starting out with it, so feel free to ask me any questions or clarify any concepts that you're unsure about. What specific areas of LangChain would you like to explore further?
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further?
Human: Can you summarize what you know about me from this conversation?
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W6D1 TASK 3: SIMPLE AGENT
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Agent Task 1: Calculate 25 * 4 + 10
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> Entering new AgentExecutor chain...
Action: calculator
Action Input: 25 * 4 + 10110Final Answer: The result of the calculation is 110.

> Finished chain.
Final Answer: The result of the calculation is 110.

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Agent Task 2: Use the web search tool to find information about LangChain
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> Entering new AgentExecutor chain...
Action: web_search_stub
Action Input: LangChainWeb search result for 'LangChain': LangChain is a framework for developing applications powered by language models.Final Answer: The final answer to the user's question is that LangChain is a framework for developing applications powered by language models.

> Finished chain.
Final Answer: The final answer to the user's question is that LangChain is a framework for developing applications powered by language models.

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> Finished chain.
Final Answer: The result of the calculation is 110.
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Agent Task 2: Use the web search tool to find information about LangChain
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> Entering new AgentExecutor chain...
Action: web_search_stub
Action Input: LangChainWeb search result for 'LangChain': LangChain is a framework for developing applications powered by language models.Final Answer: The final answer to the user's question is that LangChain is a framework for developing applications powered by language models.

> Finished chain.
Final Answer: The final answer to the user's question is that LangChain is a framework for developing applications powered by language models.
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Agent Task 3: Calculate 100 / 5 + 20
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> Entering new AgentExecutor chain...
Action: calculator
Action Input: 100 / 5 + 2040.0Final Answer: The result of the calculation is 40.0.

> Finished chain.
Final Answer: The result of the calculation is 40.0.
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W6D1 ALL TASKS COMPLETED
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PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W6D1_LangChain_Fundamentals>
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